

Ritesh Kumar

Ph.D. in Electrical Engineering | Senior Engineer, Qualcomm
Communication-Efficient Statistical Inference | Signal Processing | Computational Imaging
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Research Profile

My research focuses on distributed statistical estimation and learning under limited communication. I have worked on distributed mean estimation, quantization, one-bit communication, and estimation for log-concave distributions. I am also interested in information theory, statistical signal processing, and federated learning. At Qualcomm, I work on camera image and video quality, computational imaging, and image-quality analysis tools.

Professional Experience

Senior Engineer – Camera Image and Video Quality Jan. 2026–Present
Qualcomm

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- Conduct hardware-based camera experiments under controlled lighting conditions using standard image-quality charts and laboratory setups, including MCC, TE42, and DXOMARK-based test environments.
- Perform subjective and objective image/video quality analysis and characterize camera performance using standard and custom image-quality metrics.
- Develop MATLAB- and Python-based tools for image-quality analysis, metric computation, visualization, and automated evaluation of camera systems.
- Work with standardized image-quality evaluation procedures, including ISO-based measurements, for quantitative characterization of imaging pipelines.
- Apply statistical estimation, signal-processing, and data-analysis techniques to camera calibration, image-quality measurement, and characterization of practical imaging systems.

Education

Ph.D. in Electrical Engineering 2020–2026
Indian Institute of Technology Hyderabad, India
Thesis: *Distributed Mean Estimation under Limited Communication*
Advisor: Dr. Shashank Vatedka
CPI: 9.0/10

M.Tech. in Electrical Engineering 2018–2020
Indian Institute of Technology Patna, India
Advisor: Dr. Sudhir Kumar
CPI: 8.31/10

B.Tech. in Electronics and Communication Engineering 2013–2017
Punjab Technical University, Punjab, India
Percentage: 76.77%

Publications

Under Preparation

1. **Ritesh Kumar** and Shashank Vatedka, “Mean Estimation of Log-Concave Distributions with Limited Communication,” *under preparation*.

Peer-Reviewed Publications

1. **Ritesh Kumar** and Shashank Vatedka, “One-Bit Distributed Mean Estimation with Unknown Variance,”

Transactions on Machine Learning Research (TMLR), 2026. [Paper]

2. Nithish S. *, **Ritesh Kumar***, and Shashank Vatedka, “Unbiased Quantization of the L_1 Ball for Communication-Efficient Distributed Mean Estimation,” in *Proc. 28th International Conference on Artificial Intelligence and Statistics (AISTATS)*, PMLR, vol. 258, pp. 1270–1278, 2025. [Paper]
3. Swaraj Srivastava, **Ritesh Kumar**, Pavan Reddy M., and Abhinav Kumar, “RIS-Assisted Hybrid NOMA–OMA System with Imperfections in SIC and Phase Compensation,” in *Proc. IEEE Vehicular Technology Conference (VTC)*, 2025. [Paper]
4. **Ritesh Kumar** and Shashank Vatedka, “Communication-Constrained Distributed Mean Estimation of Log-Concave Distributions,” in *Proc. 29th National Conference on Communications (NCC)*, 2023. [Best Paper Award, Communications Track] [Paper]
5. **Ritesh Kumar**, Pritam Khan, and Sudhir Kumar, “Healthcare Data Encryption Technique Using Hybrid Cellular Automata in IoT Networks,” *Wireless Personal Communications*, Springer, 2022. [Paper]
6. **Ritesh Kumar**, Pritam Khan, and Sudhir Kumar, “A Cellular Automata-Based Healthcare Data Encryption Technique for IoT Networks,” in *Proc. IEEE INDICON*, 2019. [First Prize, M. V. Chauhan Student Paper Contest] [Paper]

*Equal contribution.

Selected Research Contributions

- **Distributed Mean Estimation under Limited Communication:** Studied the fundamental trade-off between communication budget, number of clients, and estimation accuracy in distributed mean estimation, where each client can transmit only a limited number of bits.
- **Mean Estimation for Log-Concave Distributions:** Developed a communication-efficient estimation scheme for symmetric log-concave distributions using coarse localization, fine localization, and refinement stages across different communication regimes.
- **One-Bit Mean Estimation with Unknown Variance:** Studied protocols for distributed mean estimation of symmetric log-concave scale-location families under one-bit communication constraints. Showed that one round of adaptation suffices to achieve optimal asymptotic MSE, and that non-adaptive protocols can be strictly suboptimal for several scale-location families of interest.
- **Quantization for Distributed Estimation:** Developed an unbiased quantization scheme for vectors in the L_1 ball and established order-optimal worst-case mean-squared error bounds. Applied the scheme to communication-efficient distributed mean estimation and federated learning.
- **Computational Imaging and Camera Systems:** Work on image and video quality assessment, image processing, camera calibration, computational imaging, multi-view imaging, and development of tools for analyzing practical camera systems.
- **RIS-Assisted Wireless Communication:** Studied RIS-assisted hybrid NOMA–OMA systems with imperfect successive interference cancellation and phase compensation.

Honors and Awards

- **JENESYS India–Japan Science and Technology Exchange Program**, 2023.
- **Best Paper Award, Communications Track**, National Conference on Communications (NCC), 2023.
- **First Prize, M. V. Chauhan Student Paper Contest**, IEEE INDICON, 2019.
- **First Prize, Project Expo**, Techno-Opus 2016, ISTE Student Chapter, Shaheed Bhagat Singh State Technical Campus.

Technical Skills

Programming: Python, C/C++, MATLAB

Scientific Computing and Machine Learning: PyTorch, NumPy, SciPy

Tools: Git, $\text{\LaTeX}$

Teaching Experience

Teaching Assistant, IIT Hyderabad

Courses: Linear Algebra; Information Theory; Convex Optimization; Topics in Data Storage and Communication; Wireless Communication; Statistical Learning Theory.

Responsibilities: Conducted tutorials, evaluated assignments and examinations, supported course instruction, and assisted students with coursework and projects.

Teaching Assistant, IIT Patna

Course: Digital Circuits and Design.

Responsibilities: Assisted with laboratory sessions, evaluated assignments, and supported students during laboratory and course activities.

Academic and Professional Activities

- Volunteer, IEEE Information Theory Society Summer School, IIT Hyderabad, 2024.
- Participant, Reinforcement Learning Workshop, Indian Institute of Science (IISc), Bengaluru, 2025.